

Bioinformatics Workshop

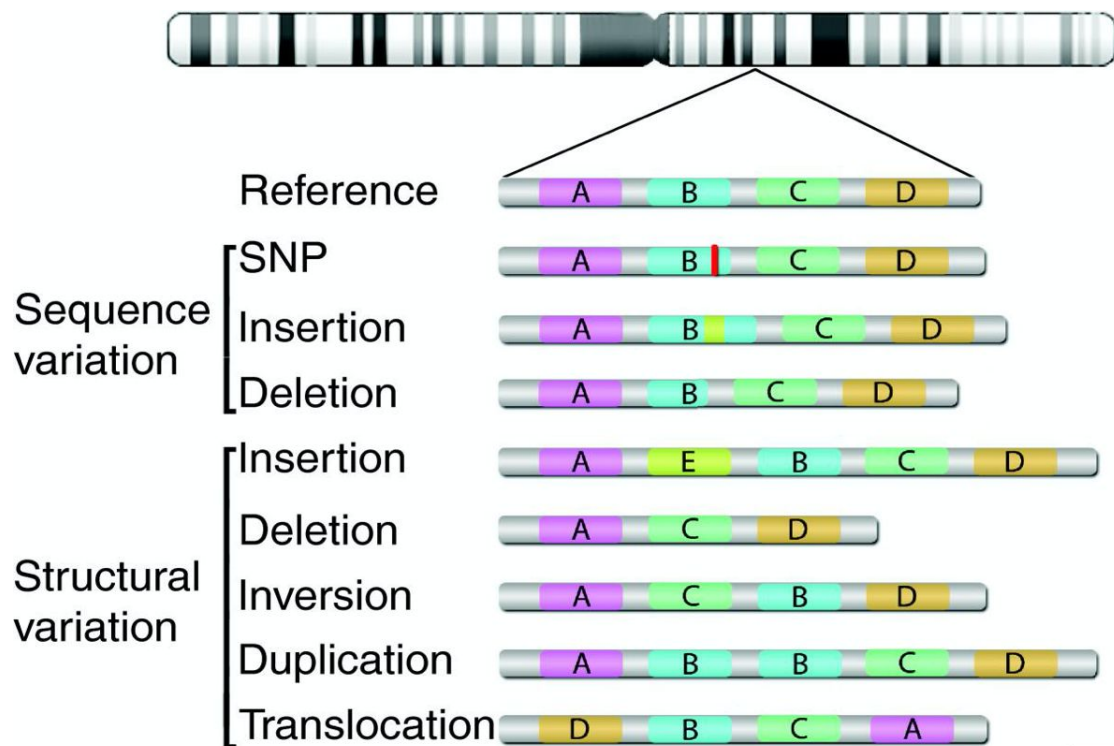
Week 06

Variant Calling

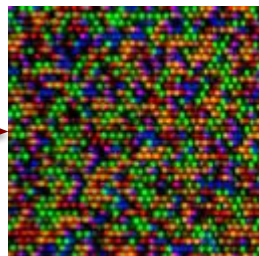
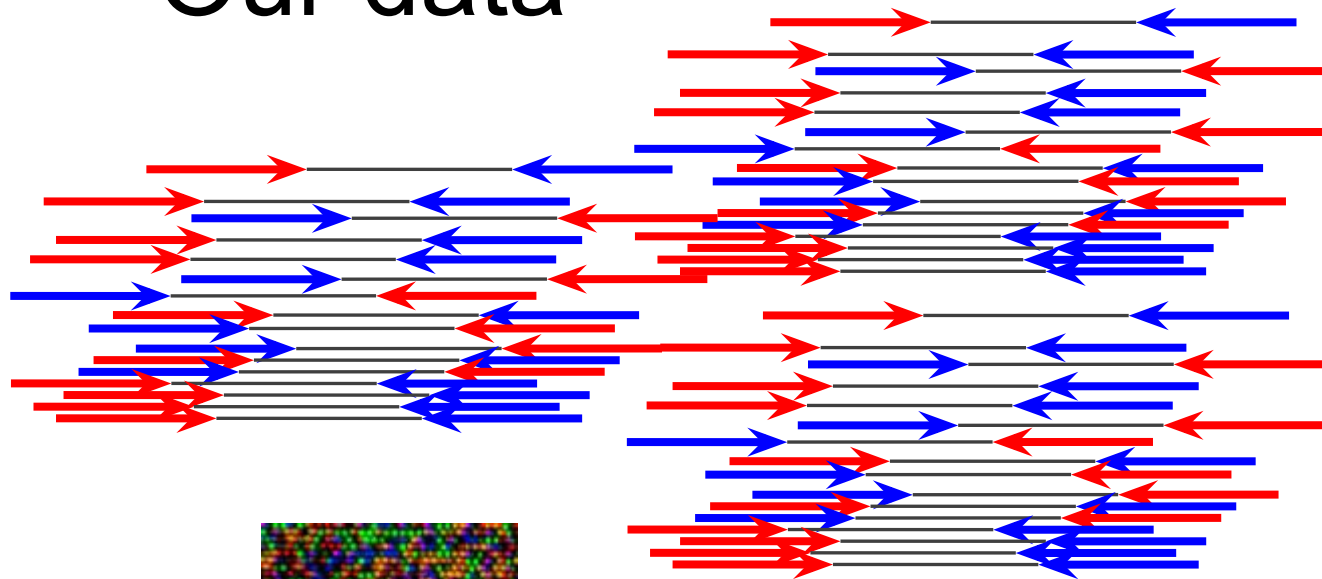
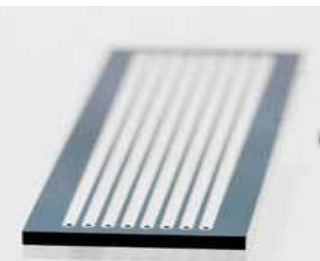
Chris Miller

Some slides adapted from Dave Larson, Aaron Quinlan, Sam Peters, and Alex Paul

Small Variant Calling

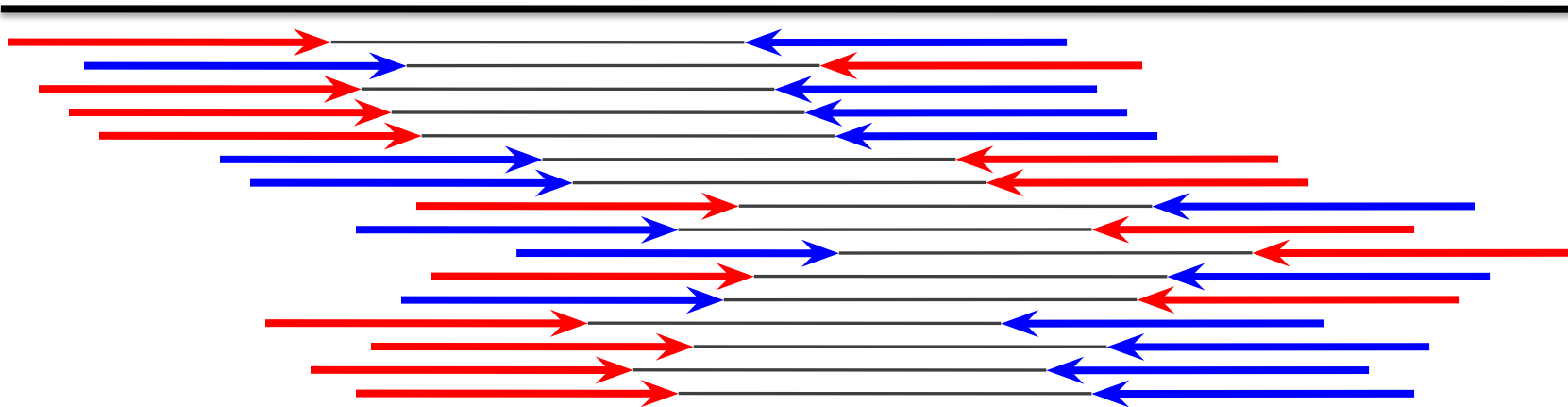


Our data



Mapping

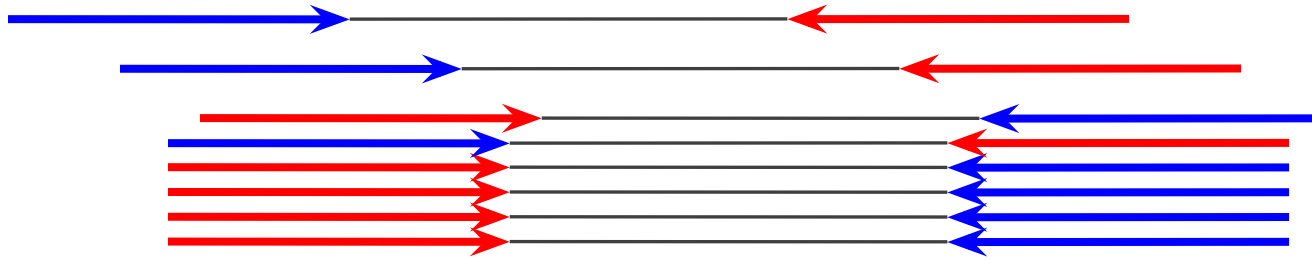
Genome Reference Sequence



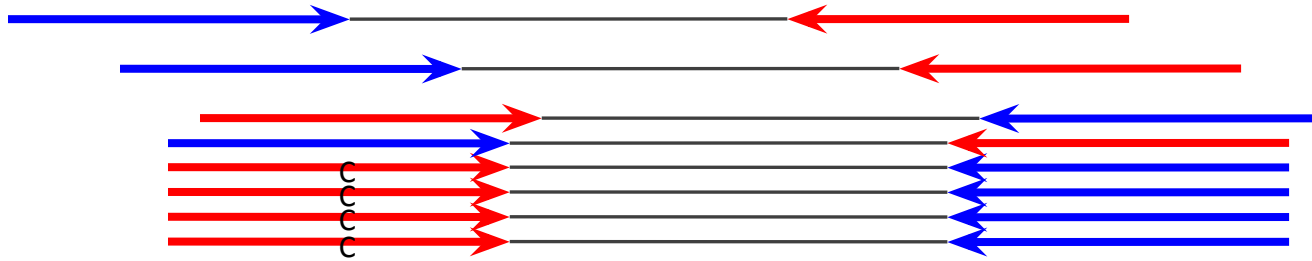
- Single-end reads can be longer, less unique depending on sequence context
- Paired-end reads can span repetitive regions, provide additional information
- Mapping has gotten quite fast, <24 hours for 120 Gbp of sequence
- Split-read alignments are the norm (BWA mem)



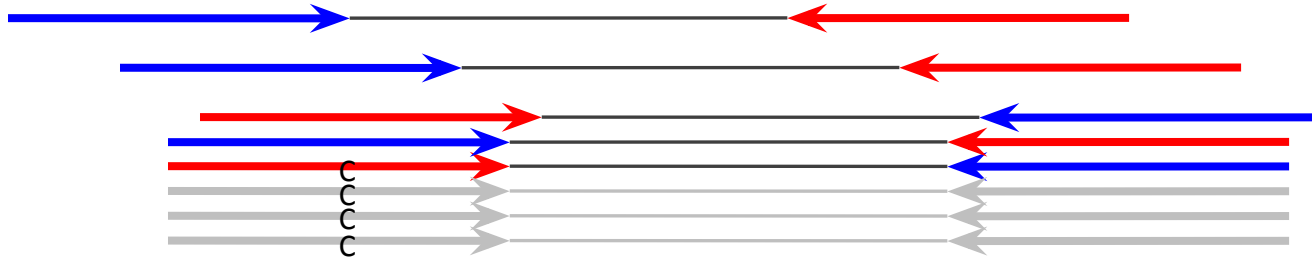
Duplication



Duplication



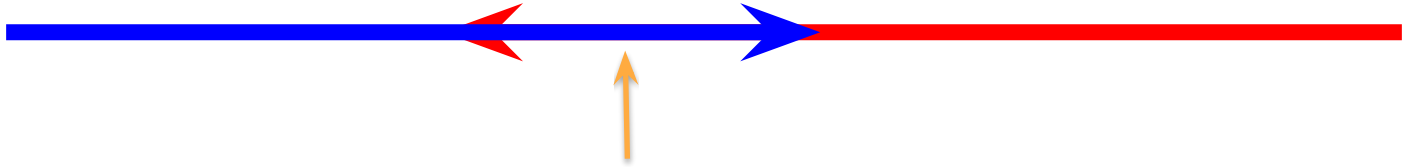
Duplication



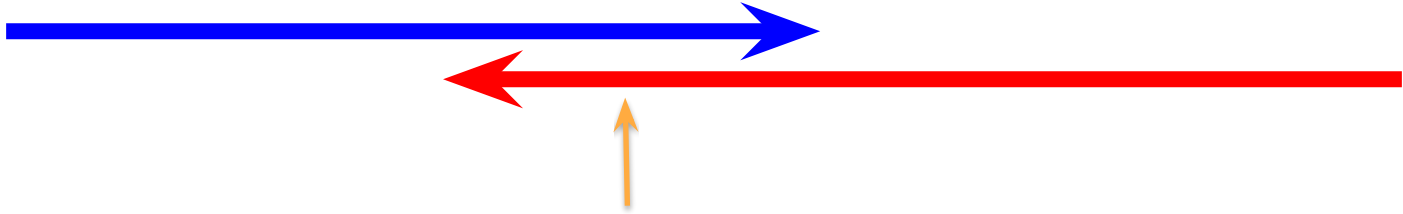
Overlapping reads



Overlapping reads



Overlapping reads



Overlapping reads



Every aspect of this process is fraught with error

- Base calling is not perfect: *0.5% error on average*
- Mapping is not perfect: *the reads are short*
- The reference sequence is not perfect

We have a little help

- Some uncertainty is encapsulated in quality scores
 - the rate at which the data is expected to be wrong
- Each base call (ACTGN) comes with a quality
 - Phred-scaled ($-10 * \log_{10}$ of quality)
 - A base call with quality of 20 is wrong 1 out of every 100 times.
- Read mapping has quality too
 - These are also Phred-scaled

Goals of a Variant Caller

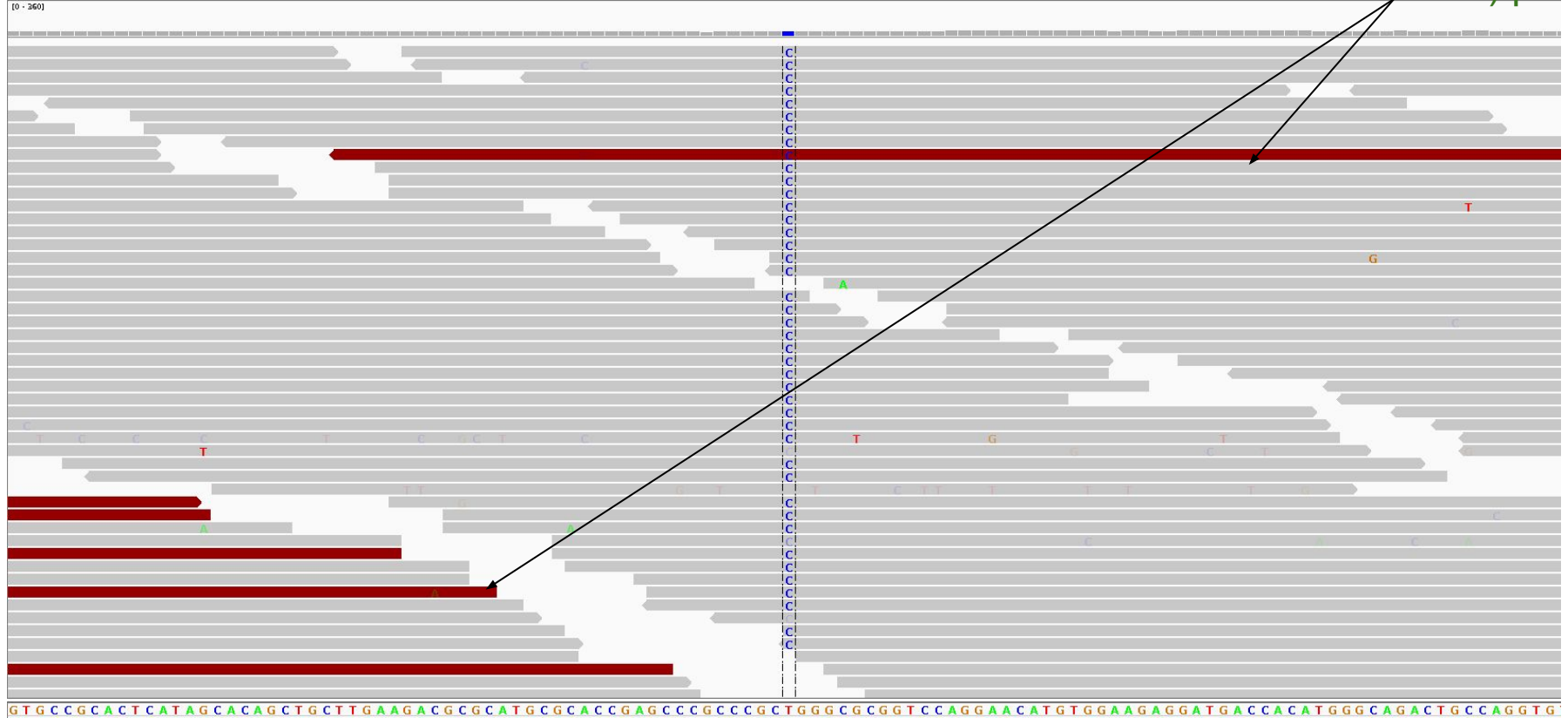
- Sensitive detect mutations
- Precisely detect mutations
 - Confounded by the error we just talked about
 - FDR must be very low as we're looking across a very large space!

Goals of a Variant Caller

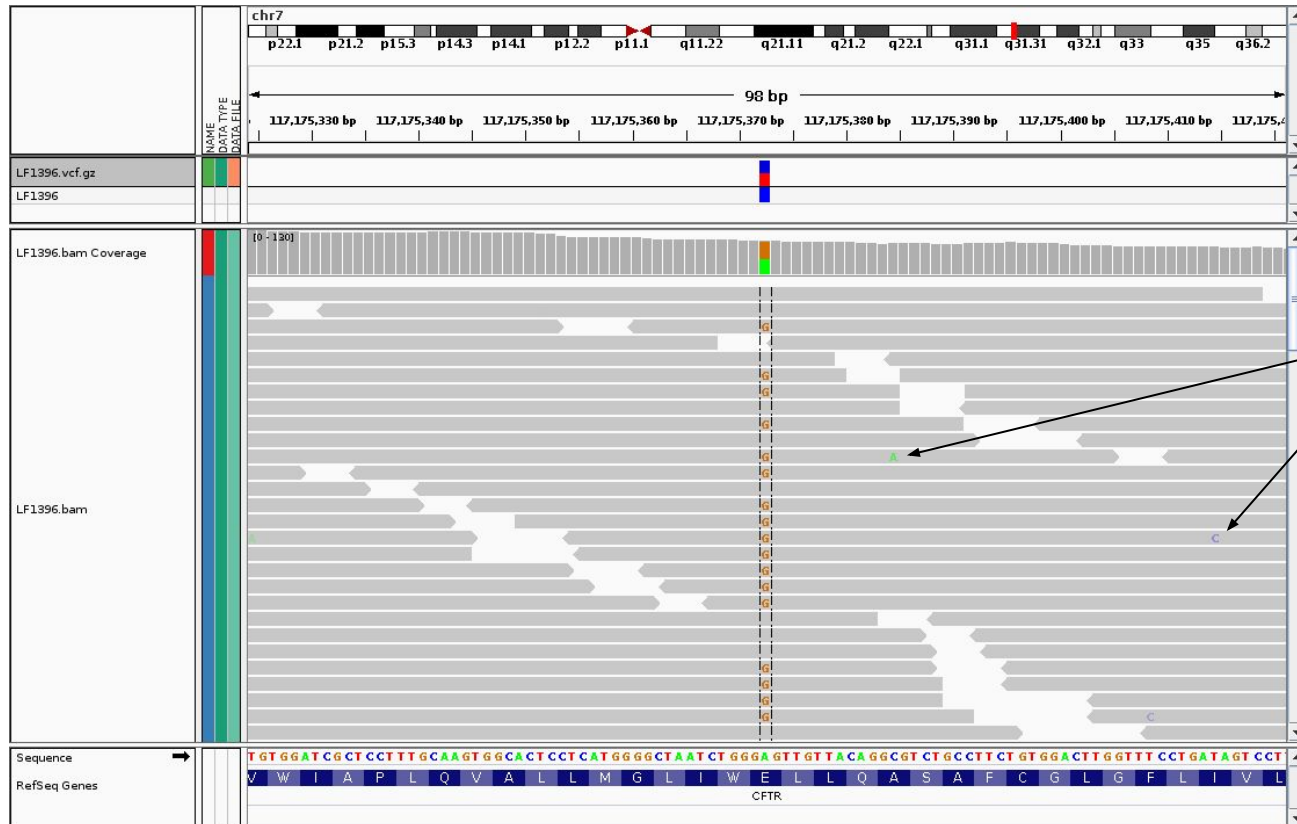
- Sensitive detect mutations
- Precisely detect mutations
 - Confounded by the error we just talked about
 - FDR must be very low as we're looking across a very large space!
 - **An FDR of 0.001 = 3.2 million false positives!**

Homozygous for the "C" allele

Improper
(too far/too
close) pairs

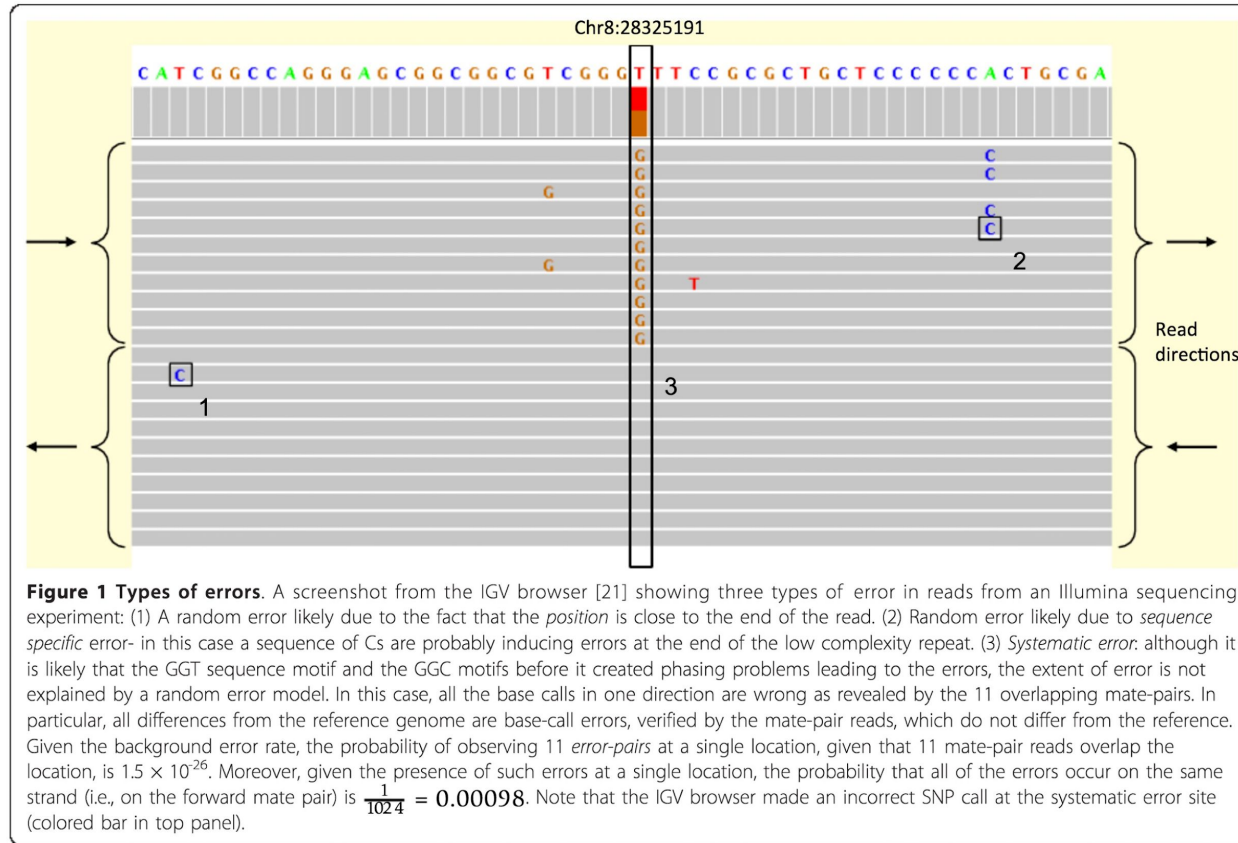


Sequencing errors fall out as noise (most of the time)

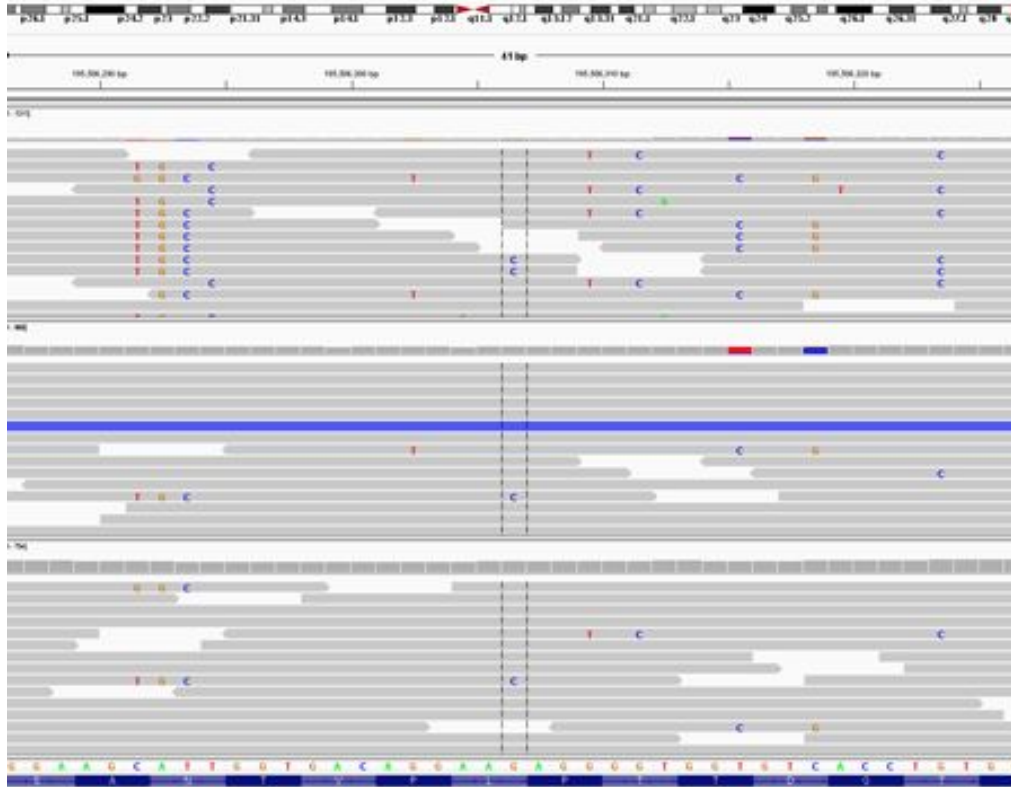


It is not always so easy

Random versus systematic error



Pileups of many differences from paralogy



RESEARCH ARTICLE | OPEN ACCESS

FLAGS, frequently mutated genes in public exomes

Casper Shyr, Maja Tarallo-Graovac, Michael Gottlieb, Jessica JY Lee, Clara van Karnebeek and Wyeth W Wasserman

BMC Medical Genomics 2014 7:64 | DOI: 10.1186/s12920-014-0064-y | © Shyr et al.; licensee BioMed Central Ltd. 2014

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Open Peer Review reports

Calling INDELs is *much* harder than SNPs

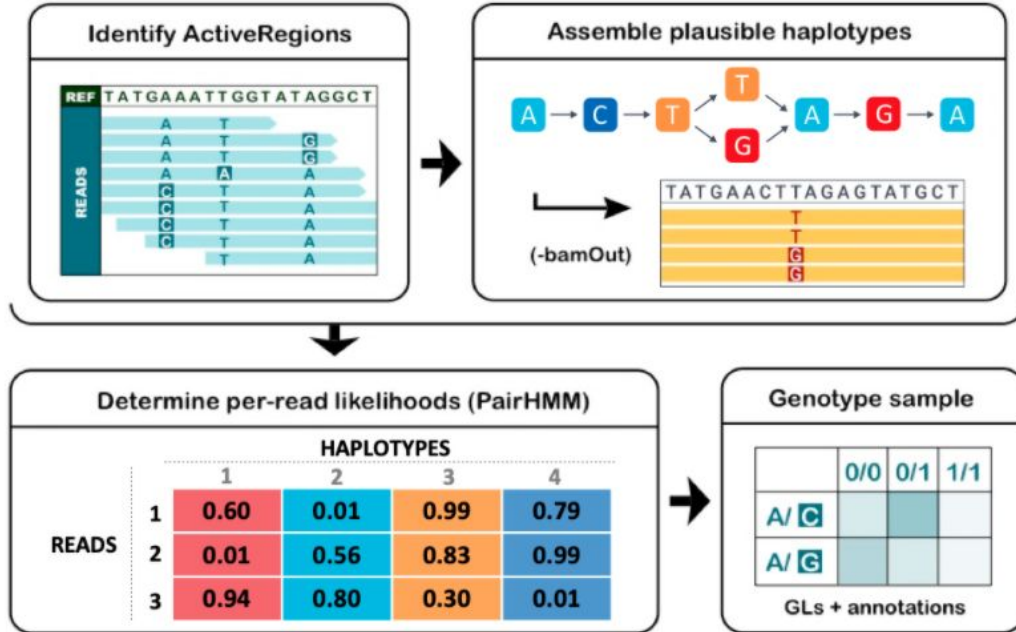


Indel "realignment"

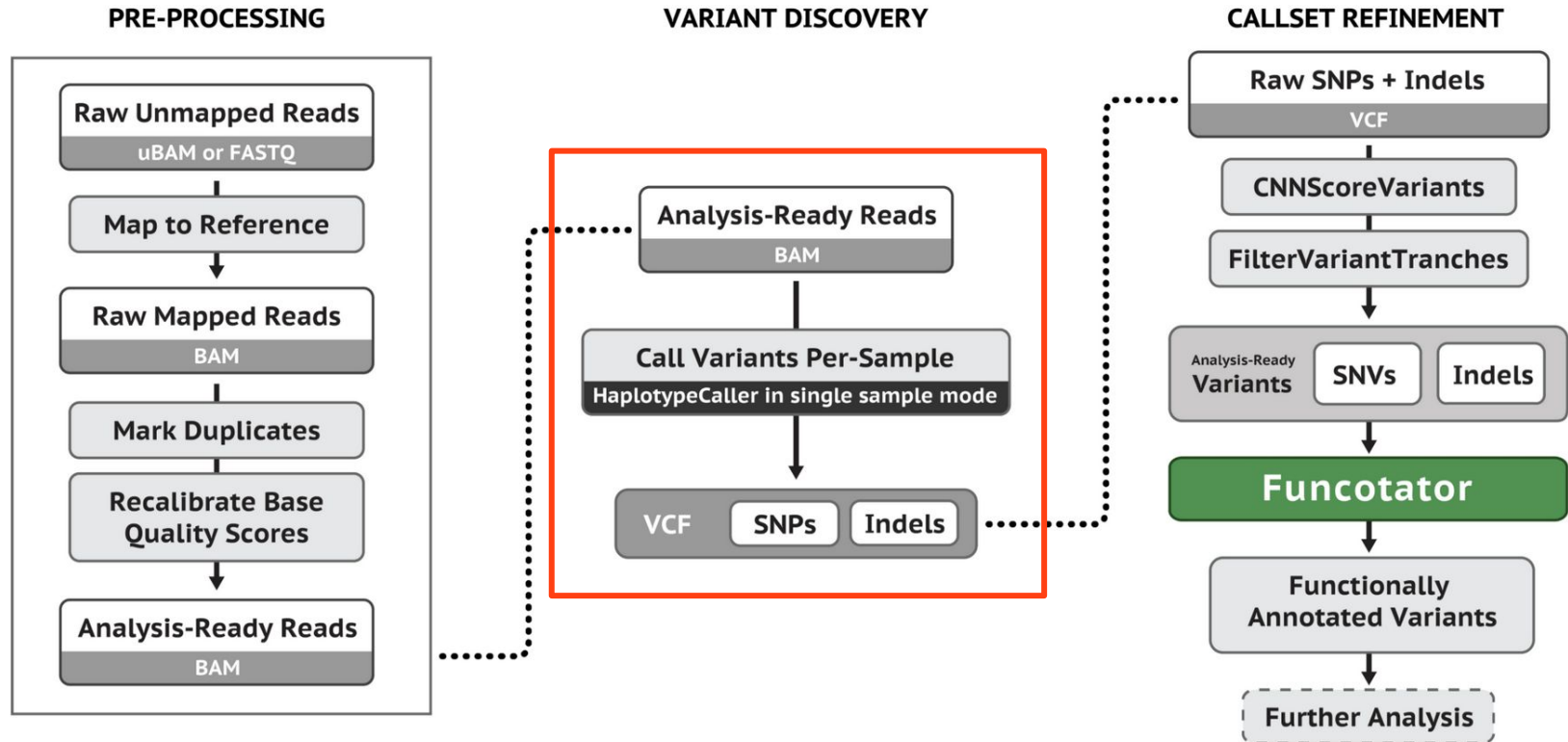


Germline SNV and Indel Calling

Call Genotypes Using GATK HaplotypeCaller



Main steps for Germline Single-Sample Data



VCF format

Sequence analysis

Advance Access publication June 7, 2011

The variant call format and VCFtools

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Associate Editor: John Quackenbush

ABSTRACT

Summary: The variant call format (VCF) is a generic format for storing DNA polymorphism data such as SNPs, insertions, deletions and structural variants, together with rich annotations. VCF is usually stored in a compressed manner and can be indexed for fast data retrieval of variants from a range of positions on the reference genome. The format was developed for the 1000 Genomes Project, and has also been adopted by other projects such as UK10K, dbSNP and the NHLBI Exome Project. VCFtools is a software suite that implements various utilities for processing VCF files, including validation, merging, comparing and also provides a general Perl API.

Availability: <http://vcftools.sourceforge.net>

Contact: rd@sanger.ac.uk

Although generic feature format (GFF) has recently been extended to standardize storage of variant information in genome variant format (GVF) (Reese *et al.*, 2010), this is not tailored for storing information across many samples. We have designed the VCF format to be scalable so as to encompass millions of sites with genotype data and annotations from thousands of samples. We have adopted a textual encoding, with complementary indexing, to allow easy generation of the files while maintaining fast data access. In this article, we present an overview of the VCF and briefly introduce the companion VCFtools software package. A detailed format specification and the complete documentation of VCFtools are available at the VCFtools web site.

VCF format

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

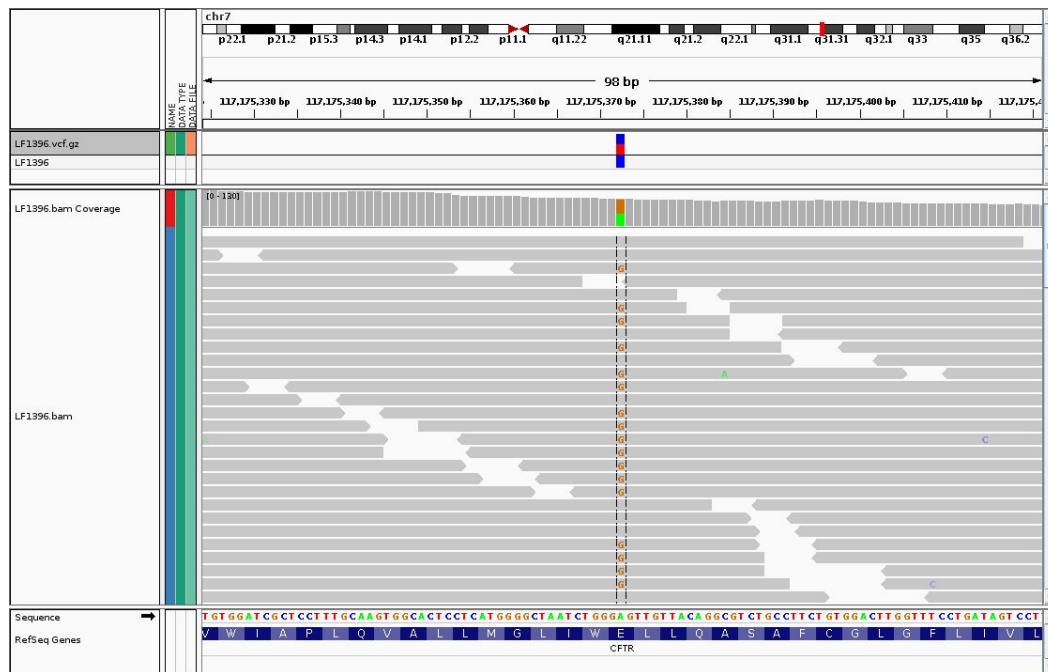
Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0/1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1/0:77	1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Annotations:

- Mandatory header lines:** ##fileformat=VCFv4.0
- Optional header lines (meta-data about the annotations in the VCF body):** ##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
- Reference alleles (GT=0):** A, T, G, C (from REF column)
- Alternate alleles (GT>0 is an index to the ALT column):** AT, CT, G, (from ALT column)
- Deletion:**
- SNP:** rs1
- Large SV:**
- Insertion:** AT, CT
- Other event:** SVTYPE=DEL;END=300
- Phased data (G and C above are on the same chromosome):** 1/1:12:3, 0/0:20

VCF format. A basic example



Heterozygous A/G.
REF allele is allele "0",
ALT is allele "1"

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	LF1396
chr7	117175373	.	A	G	90	PASS	AF=0.5	GT	0/1

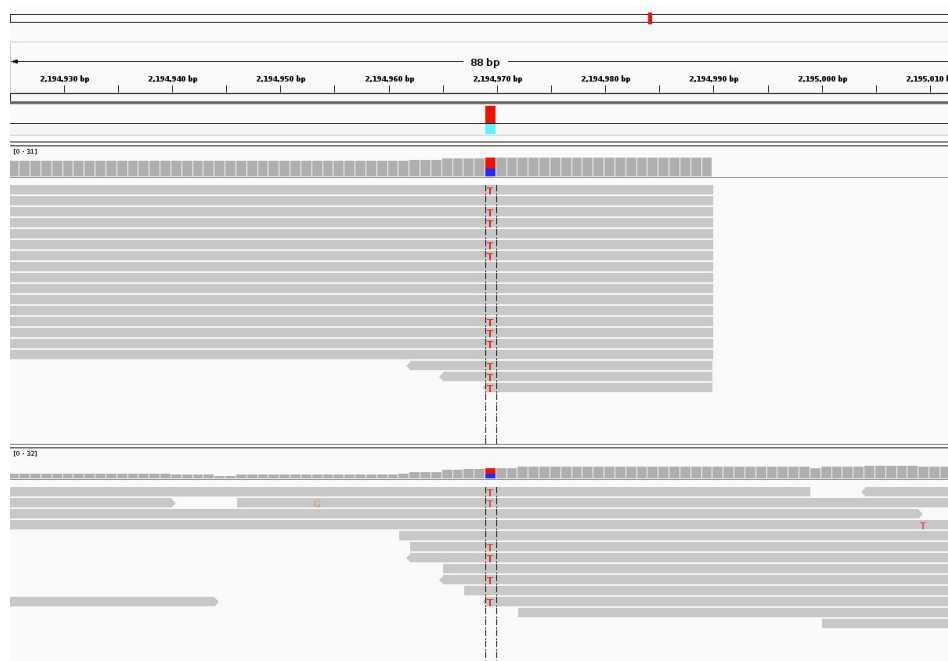
Genotypes

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	LF1396	
chr7	117175373	.	A	G	90	PASS	AF=0.0	GT	0/0	Hom. Ref.
chr7	117175373	.	A	G	90	PASS	AF=0.5	GT	0/1	Het.
chr7	117175373	.	A	G	90	PASS	AF=1.0	GT	1/1	Hom. Alt.
chr7	117175373	.	A	G	0	PASS	AF=0.0	GT	./.	Unknown

Multi-sample VCF

Mom

Kid



Heterozygous C/T

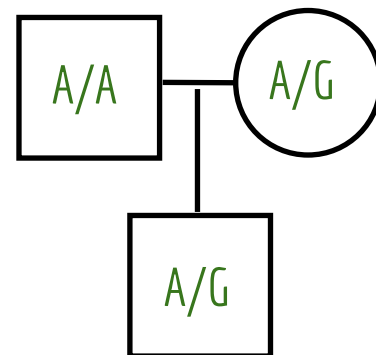
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	MOM	KID
chr7	2194169	.	C	T	210	PASS	AF=0.5	GT	0/1	0/1

Multi-sample VCF format example

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=variantcallerXYZ
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
```

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO
20	14370	rs6054257	G	A	29	PASS	NS=3;DP=14;AF=0.5;DB;H2
20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017
20	1110696	rs6040355	A	G,T	67	PASS	NS=2;DP=10;AF=0.333,0.667;AA=T;DB
20	1234567	microsat1	GTC	G,GTCT	50	PASS	NS=3;DP=9;AA=G

FORMAT	MOM	DAD	KID
GT:GQ:DP:HQ	0 0:48:1:51,51	1 0:48:8:51,51	1/1:43:5:.,.
GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3	0/0:41:3
GT:GQ:DP:HQ	1 2:21:6:23,27	2 1:2:0:18,2	2/2:35:4
GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3



"Phased" genotypes

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	MOM	KID
chr7	2194169	.	C	T	210	PASS	AF=0.5	GT	0/1	0/1

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	MOM	KID
chr7	2194169	.	C	T	210	PASS	AF=0.5	GT	0 1	0 1

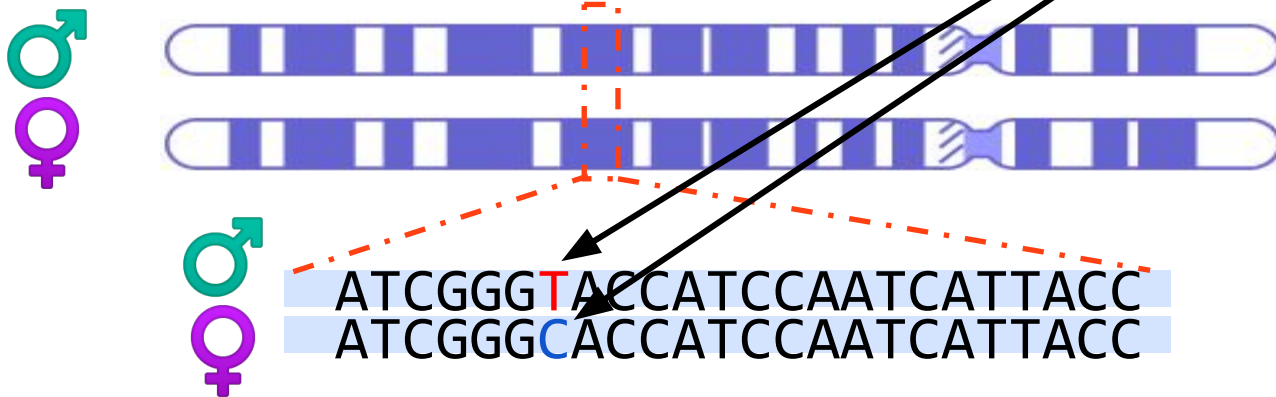
Phased genotype



"Phased" genotypes distinguish haplotypes

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	MOM	KID
chr7	2194169	.	C	T	210	PASS	AF=0.5	GT	0 1	0 1

Phased genotype



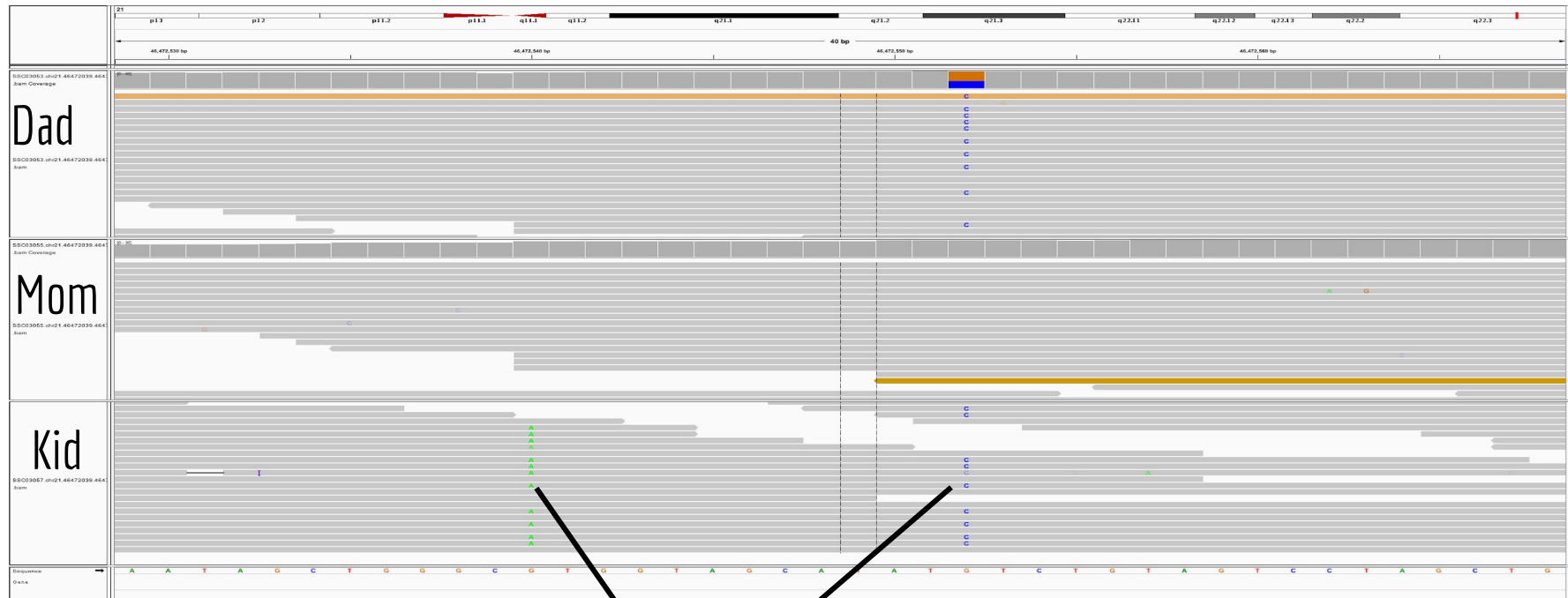
Phasing by inheritance



#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	MOM	DAD	KID
chr1	100000	.	A	G	95	PASS	AF=0.33	GT	0/1	0/0	1 0
chr1	200000	.	T	C	99	PASS	AF=0.33	GT	0/0	0/1	0 1

This is an example of a compound heterozygote. Hets where the alt allele comes from different parents at two different loci

Read-backed phasing: evidence of haplotype in read or pair



De novo mutation in the kid is linked to dad's haplotype

Next Step: Filter Raw (Unfiltered) VCF file

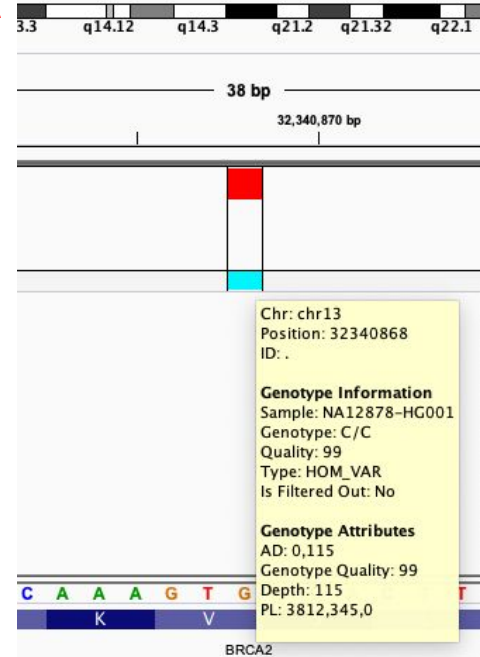
- **Problem: Too many Variants**

- Non-related Human's differ by 0.1% --> 3 Million SNPs (haploid genome)

- **Solution: Filter Low Quality and Common Variants**

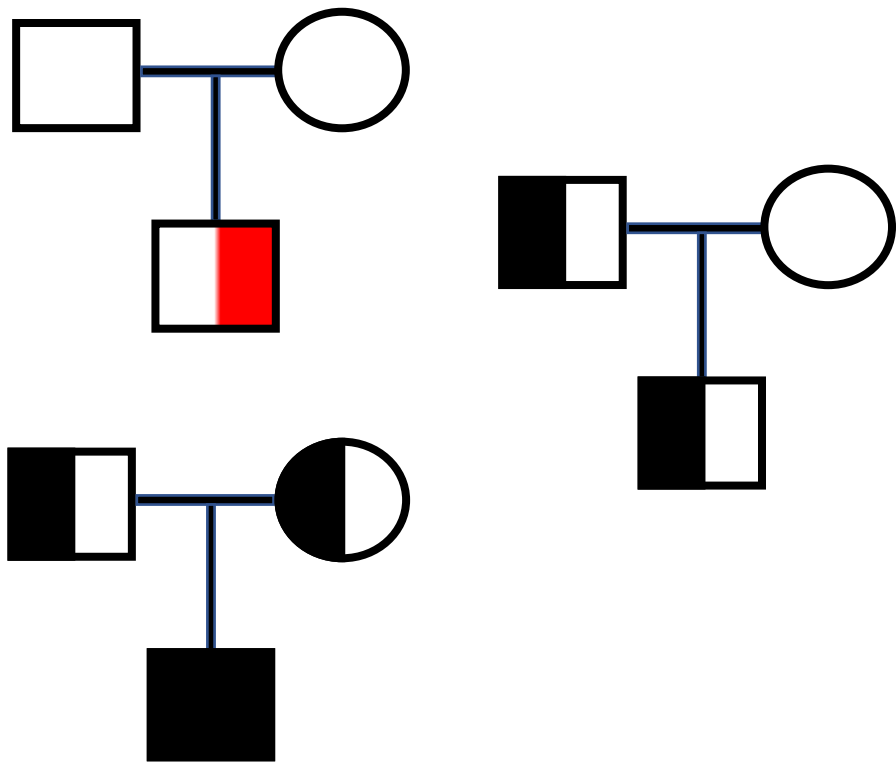
- Common Variants > 5%
- Low Quality
 - $GQ < 20$, $DP < 8$, $MQ < 40$ or GATK VQSR

VCF Visualization with IGV



Further Analysis after SNV Calling

- De-novo analysis with trios
- Rare transmitted analysis
- Compound het analysis



Somatic Mutation Calling

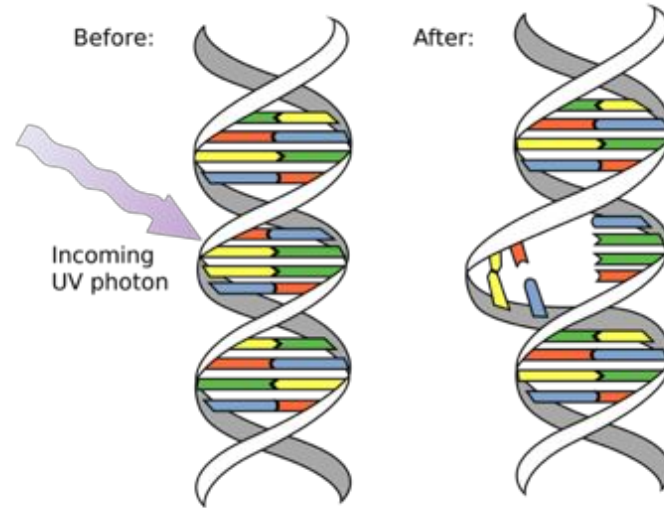
Cancer is a disease of the genome

- Cancer is caused by **somatic** mutations

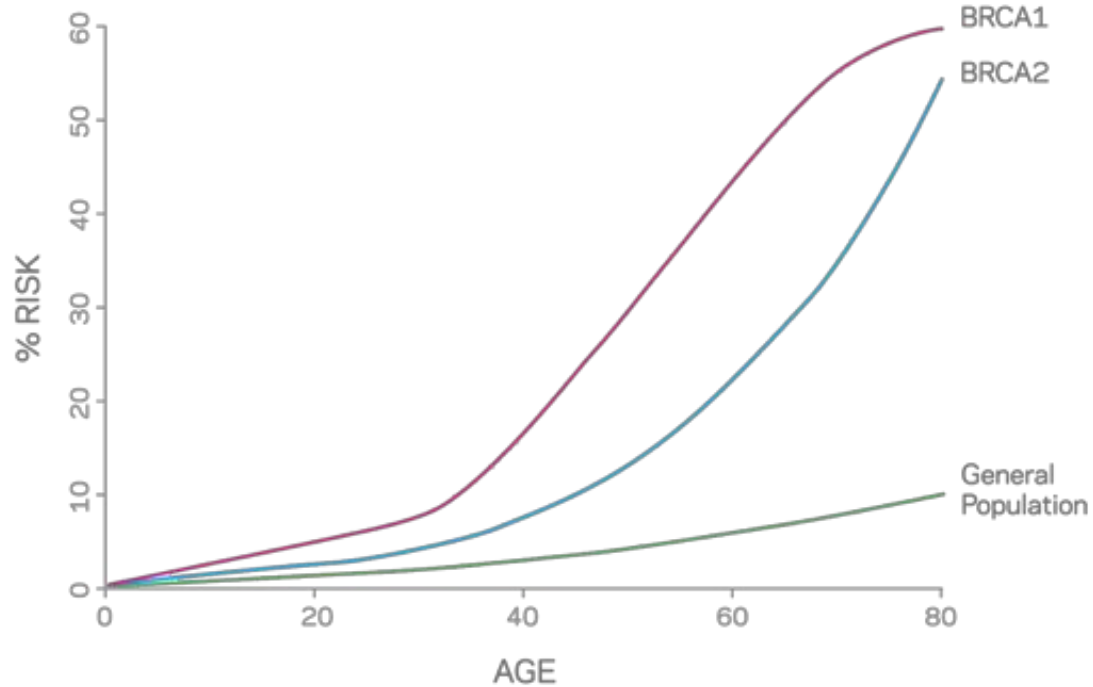


Cancer is a disease of the genome

- Cancer is caused by **somatic** mutations
- These mutations are introduced into the genome of a cell (errors in DNA copying, UV light, chemicals)
- Most cancers require around 3 driver mutations



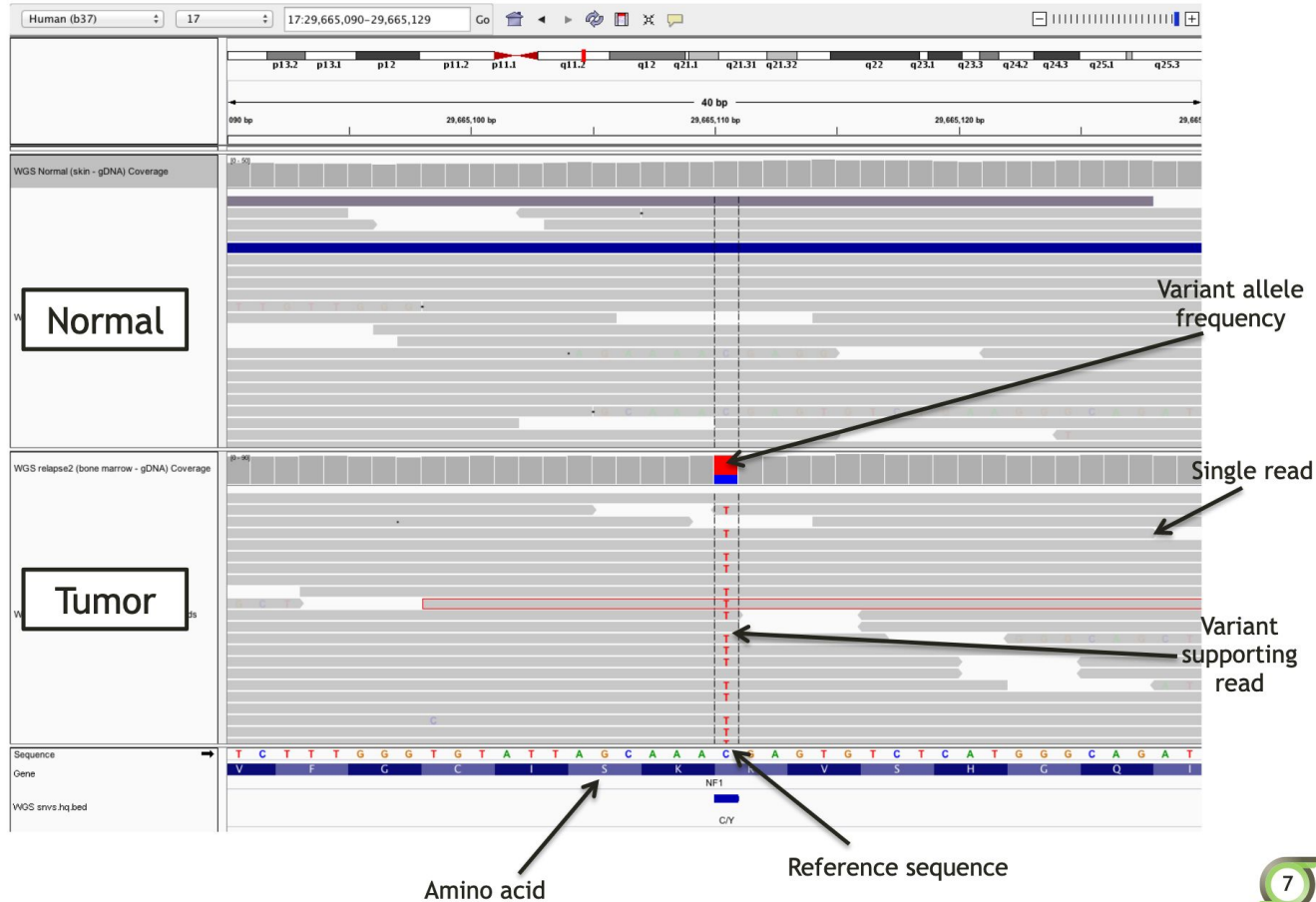
Germline Predisposition



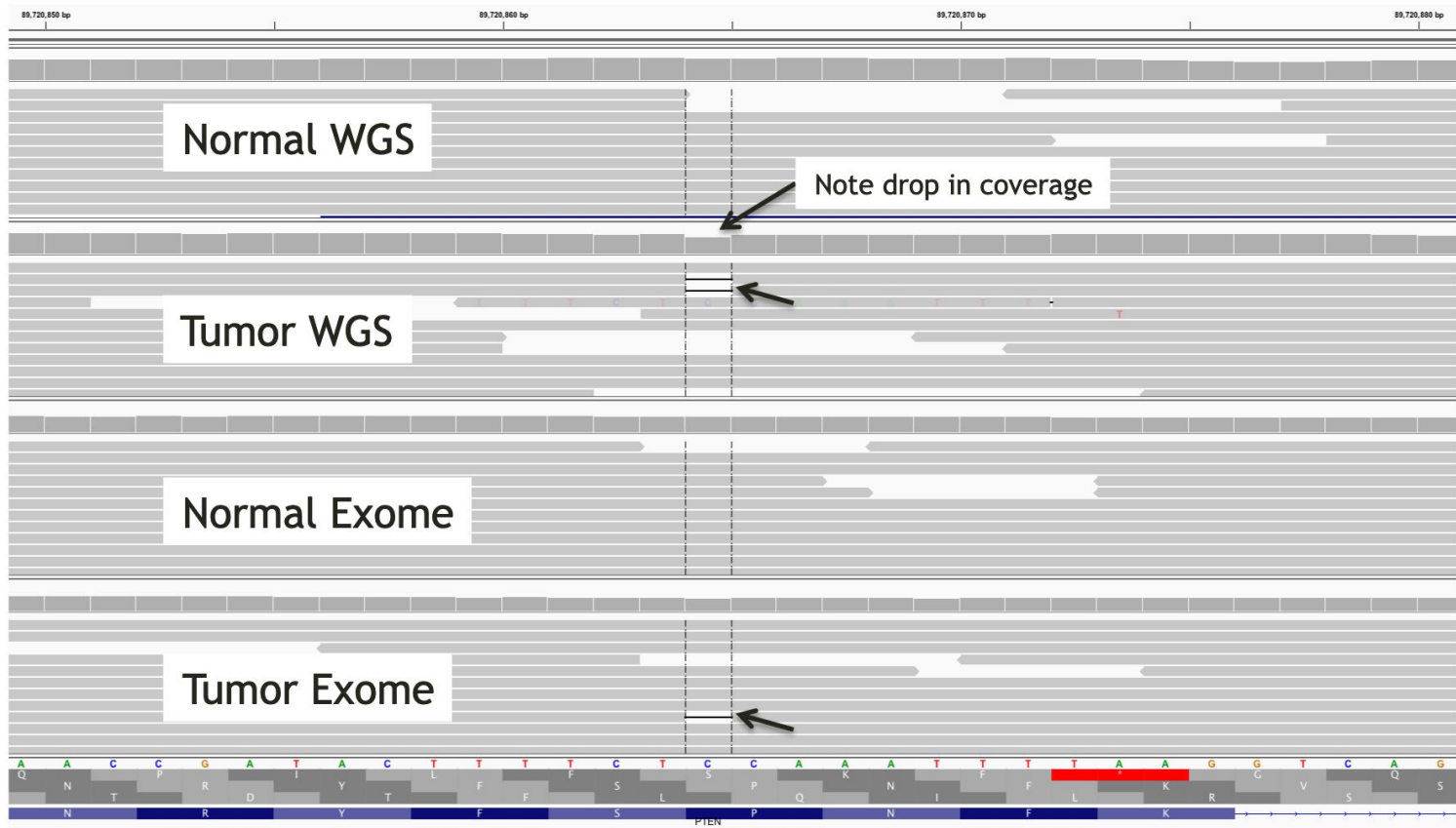
Cancer Sequencing

- In cancer, we have to (at least) double sequencing costs
- Uses both a tumor sample and a matched normal
- We compare them to find somatic mutations

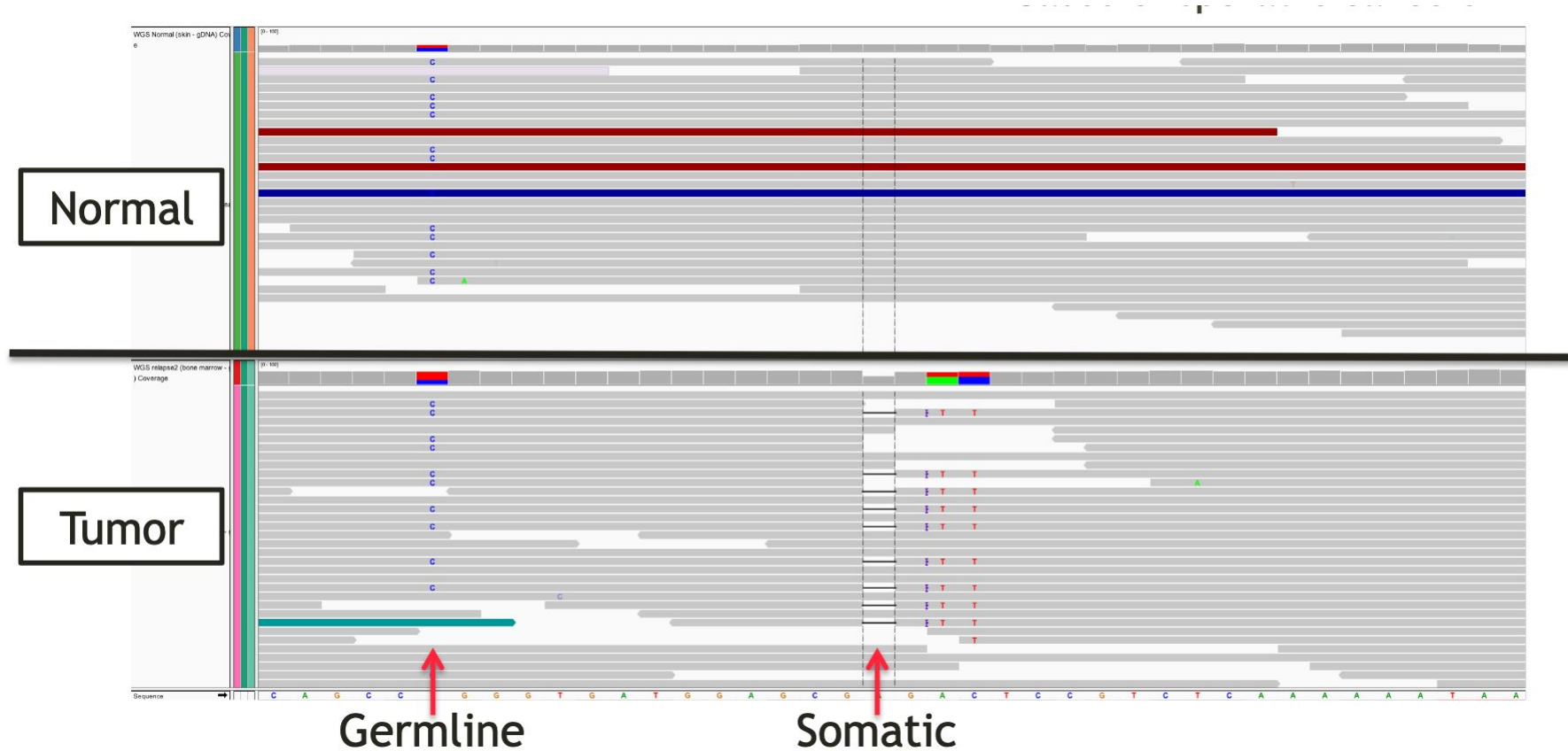
What do somatic variants look like?



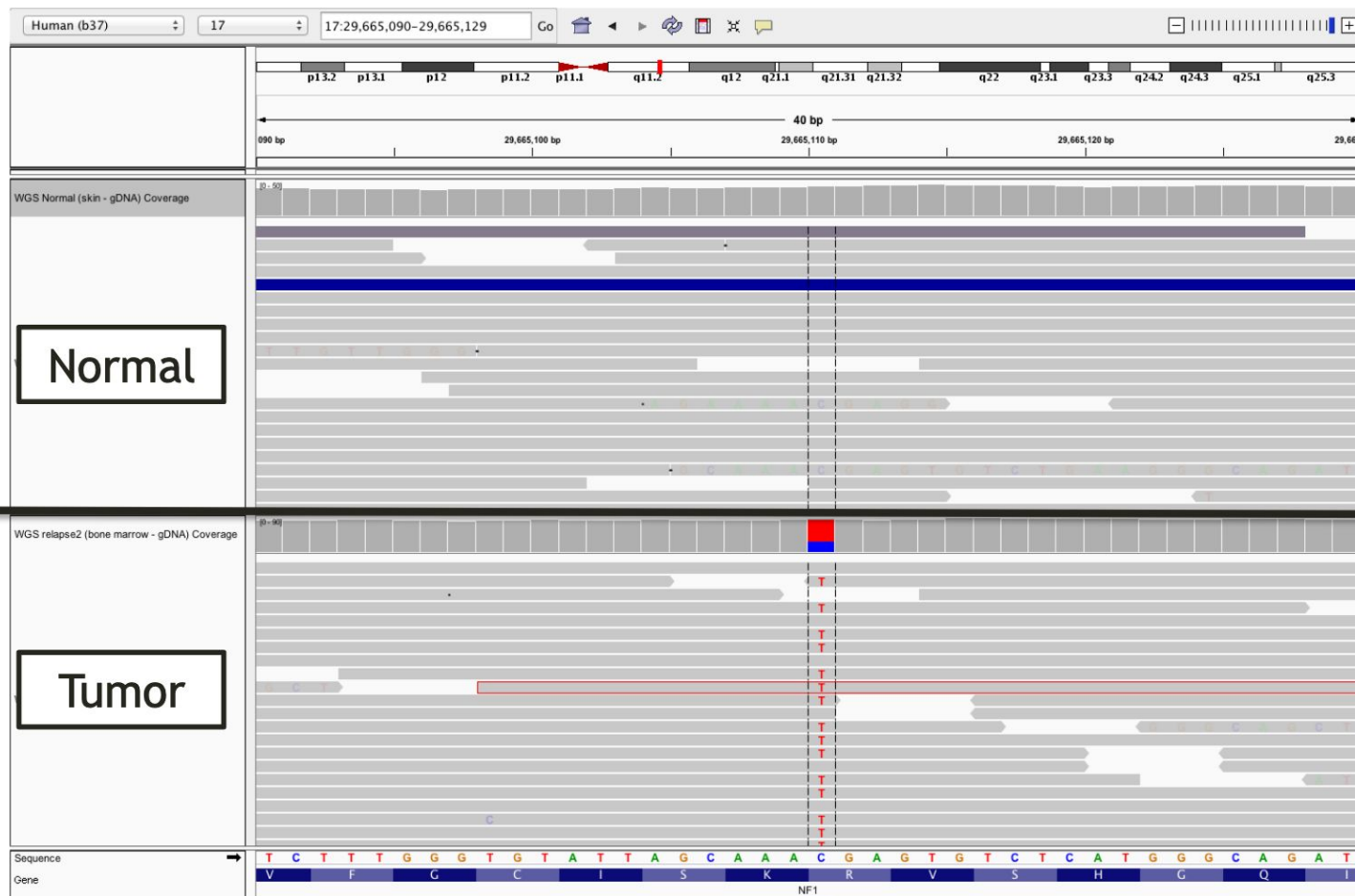
Indels



Germline vs Somatic



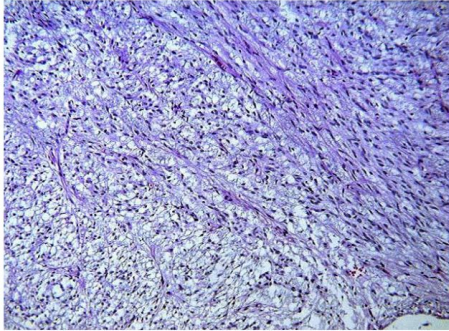
VAF = Variant reads / Total reads



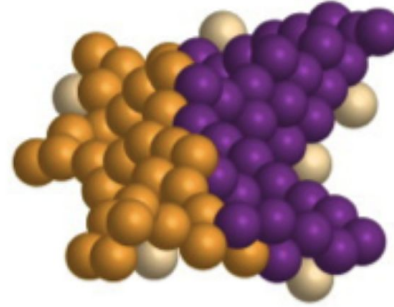
$$\text{VAF} = 0/20 = 0\%$$

$$\text{VAF} = 14/20 = 70\%$$

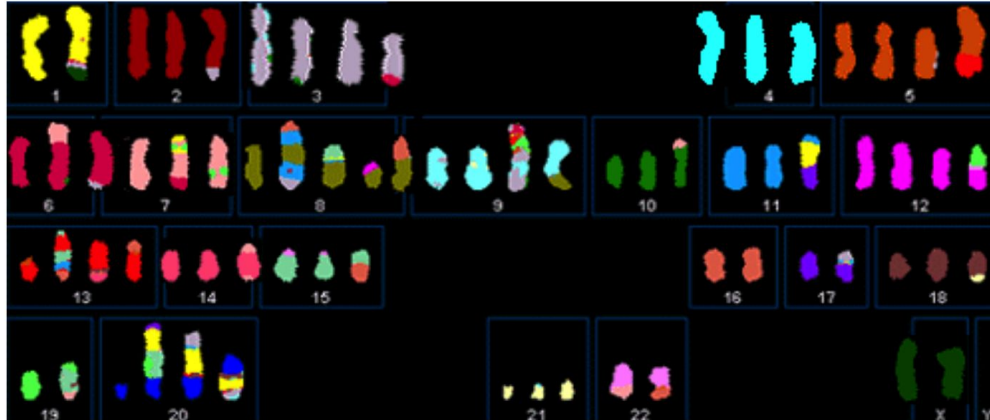
Tumors are often impure, heterogeneous, and aneuploid



Tumors are often impure
(contain normal cells)



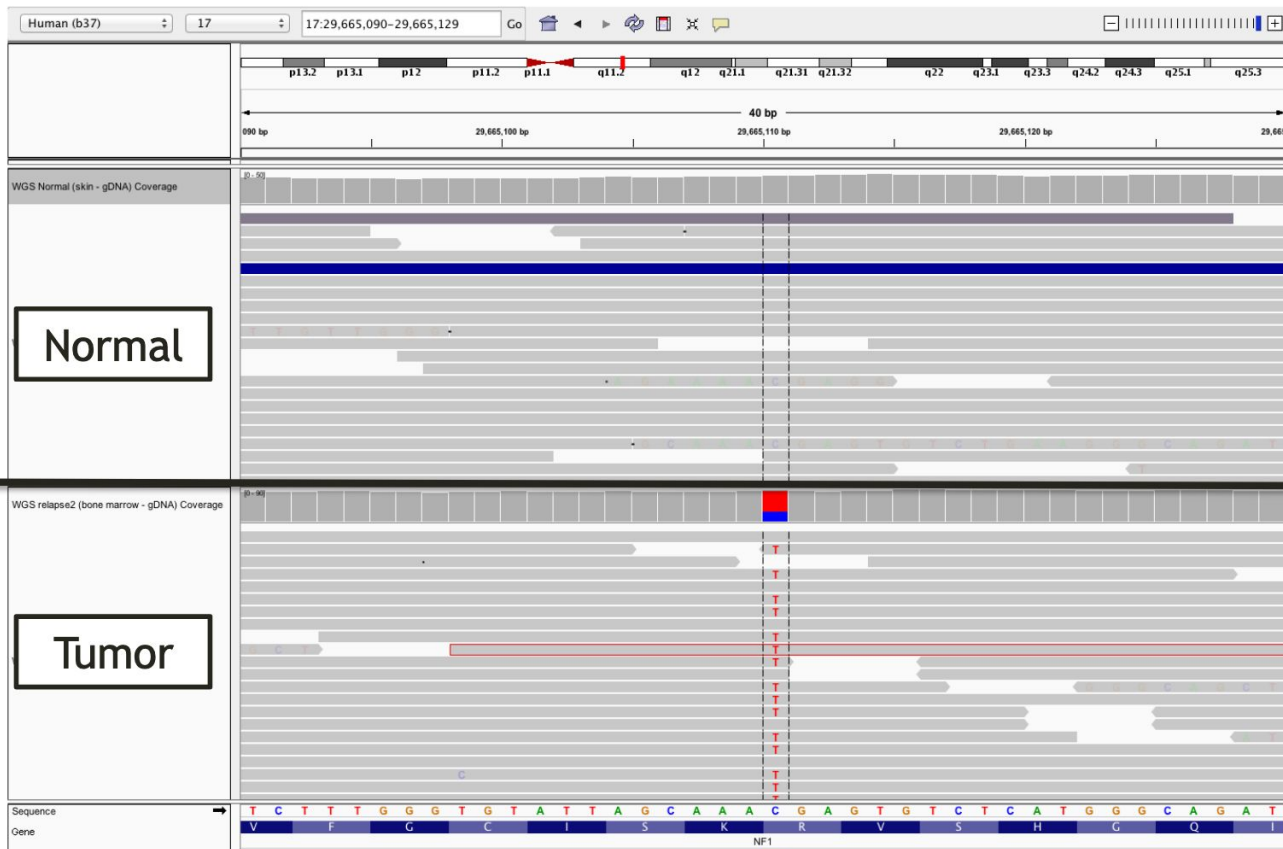
Tumors are often genetically
diverse collections of cells



Tumors may be aneuploid

How does purity influence VAF?

VAF = Variant reads / Total reads

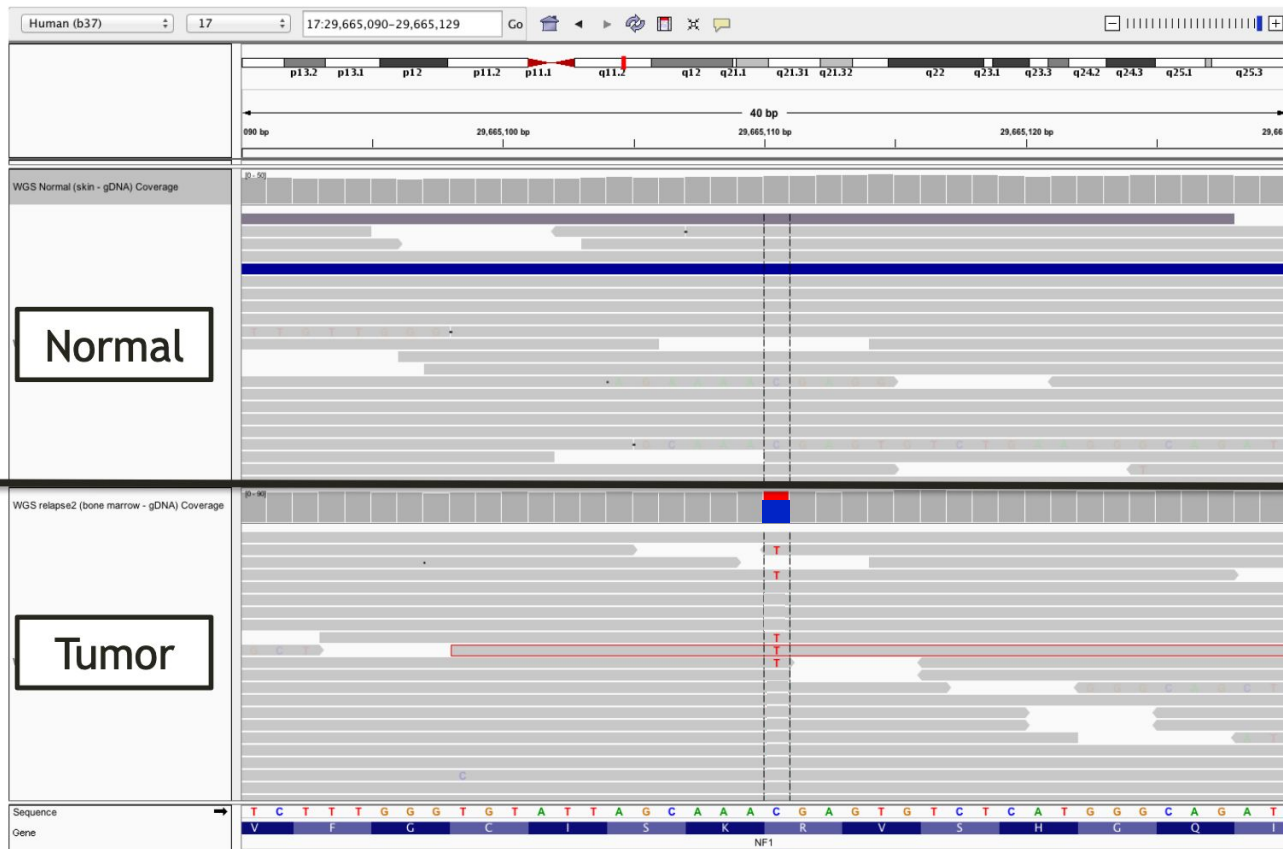


$$\text{VAF} = 0/20 = 0\%$$

$$\text{VAF} = 14/20 = 70\%$$

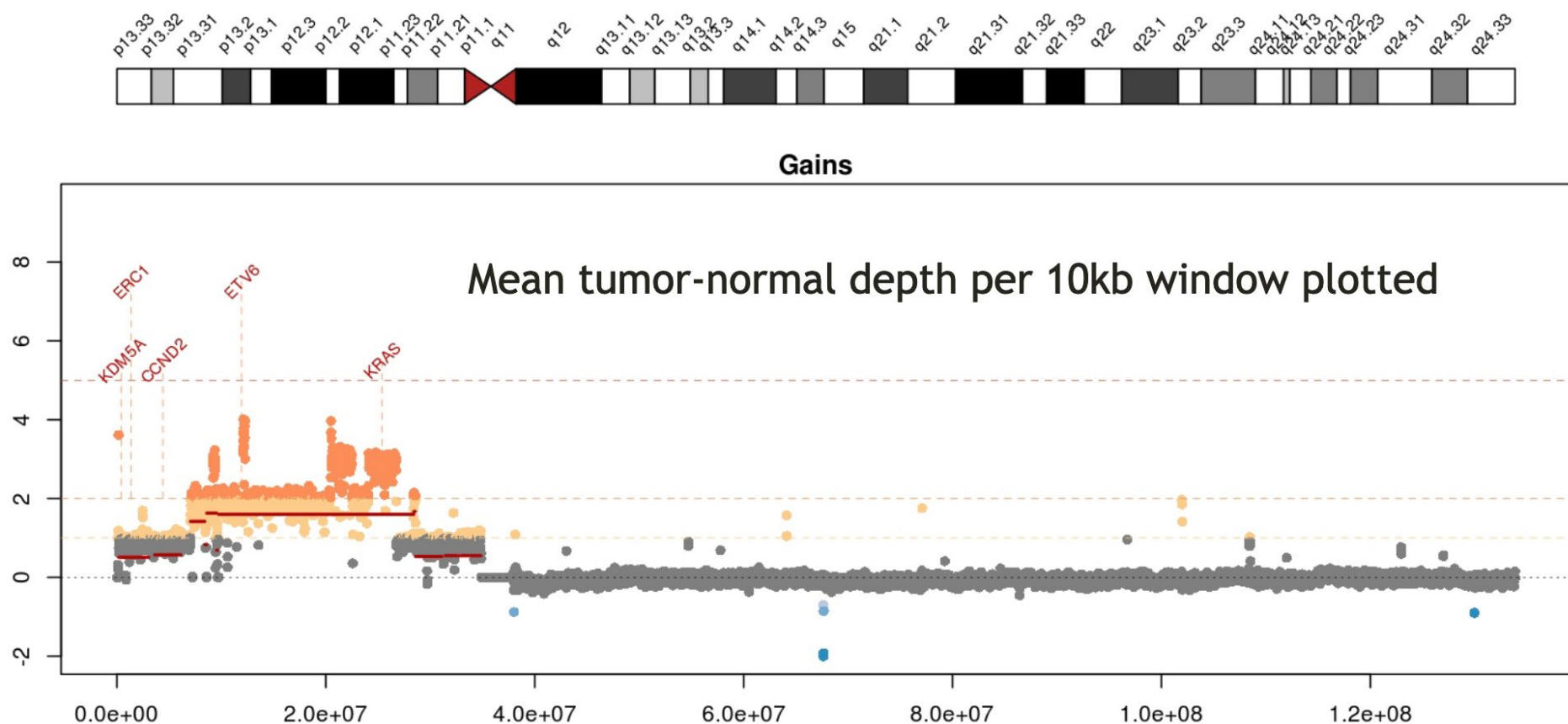
How does purity influence VAF?

VAF = Variant reads / Total reads



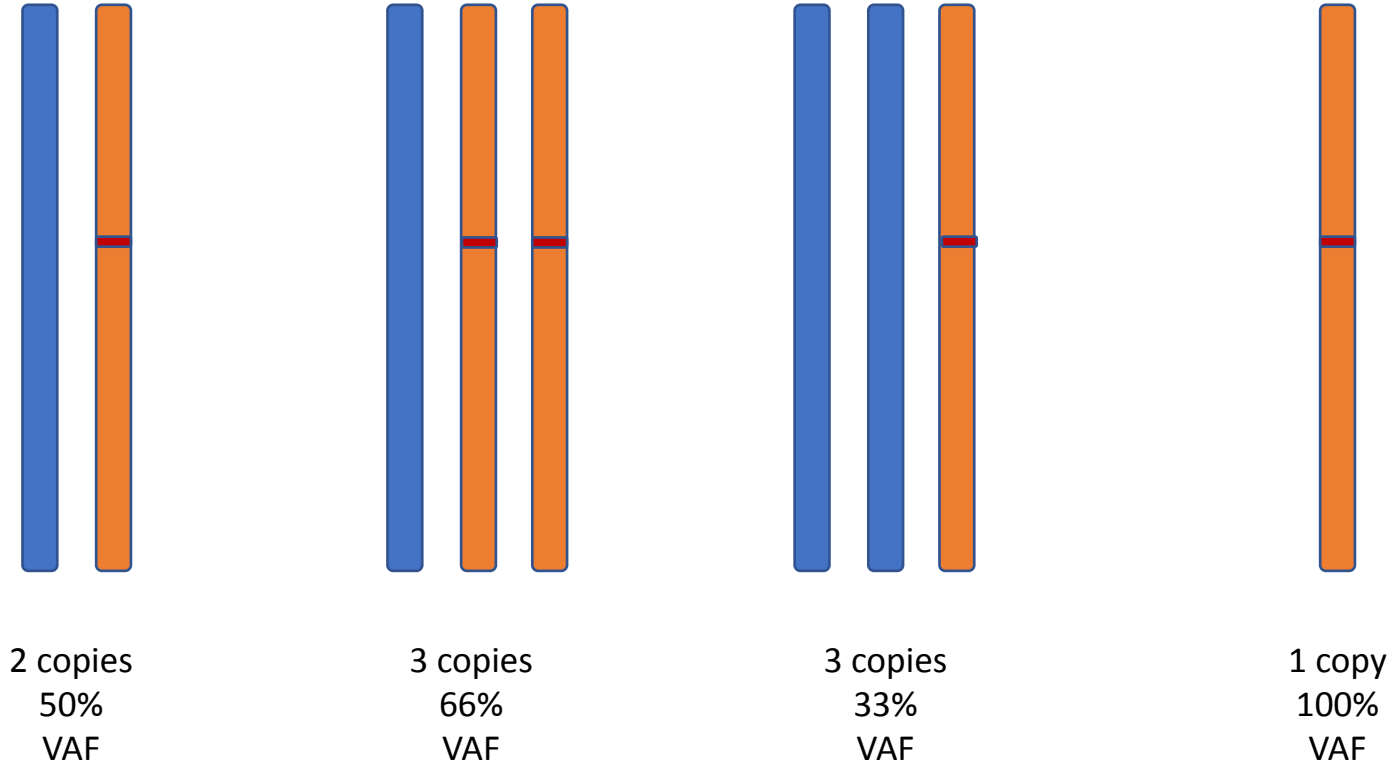
$$\text{VAF} = 0/20 = 0\%$$

$$\text{VAF} = 5/20 = 25\%$$

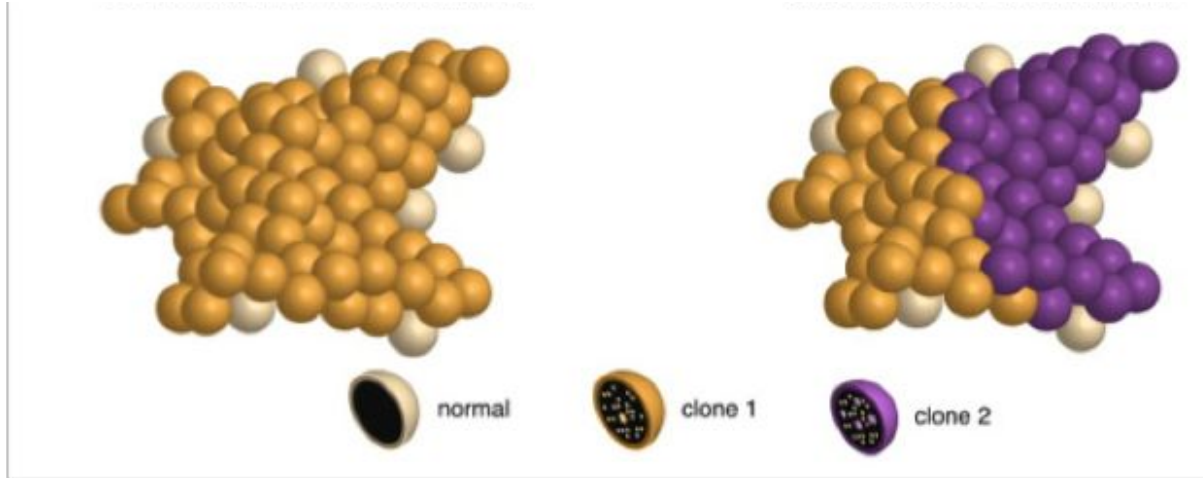


KRAS amplification in a metastatic breast cancer

How does copy number influence VAF?

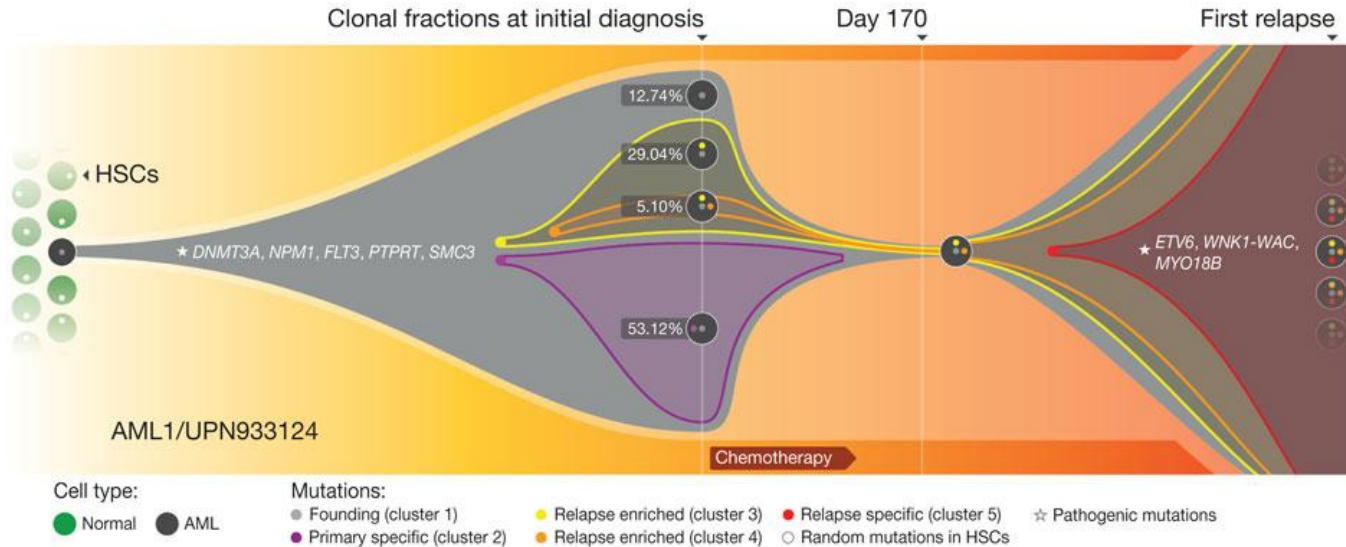


How does clonality influence VAF?

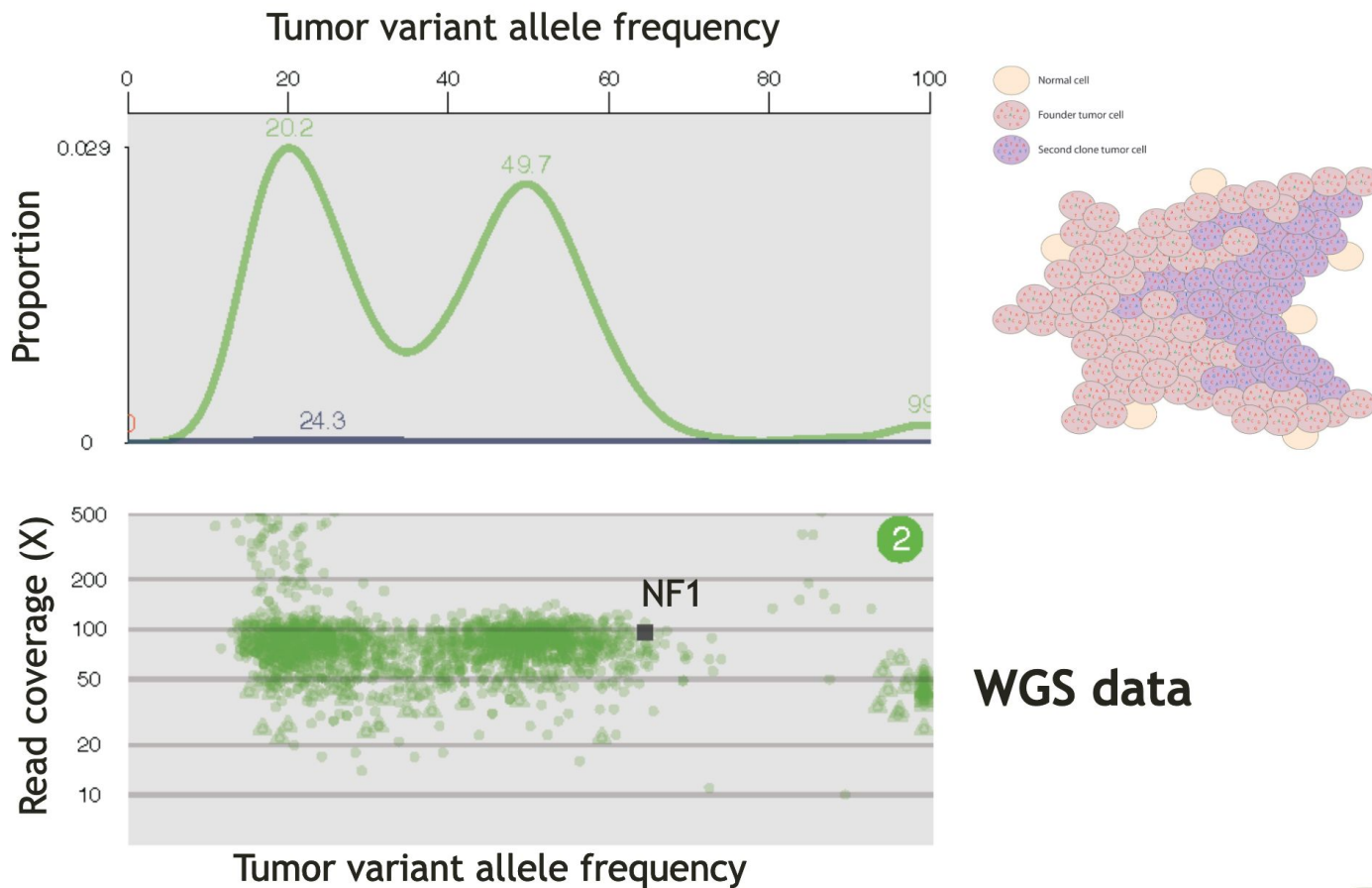


- Subclones contain genetically diverse populations of cells
- Evolution occurs at the molecular and cellular levels
- The growth rates for subclones are often different

Clonal evolution in relapsed AML



Dominant clone vs. sub-clonal (and driver vs. passenger)

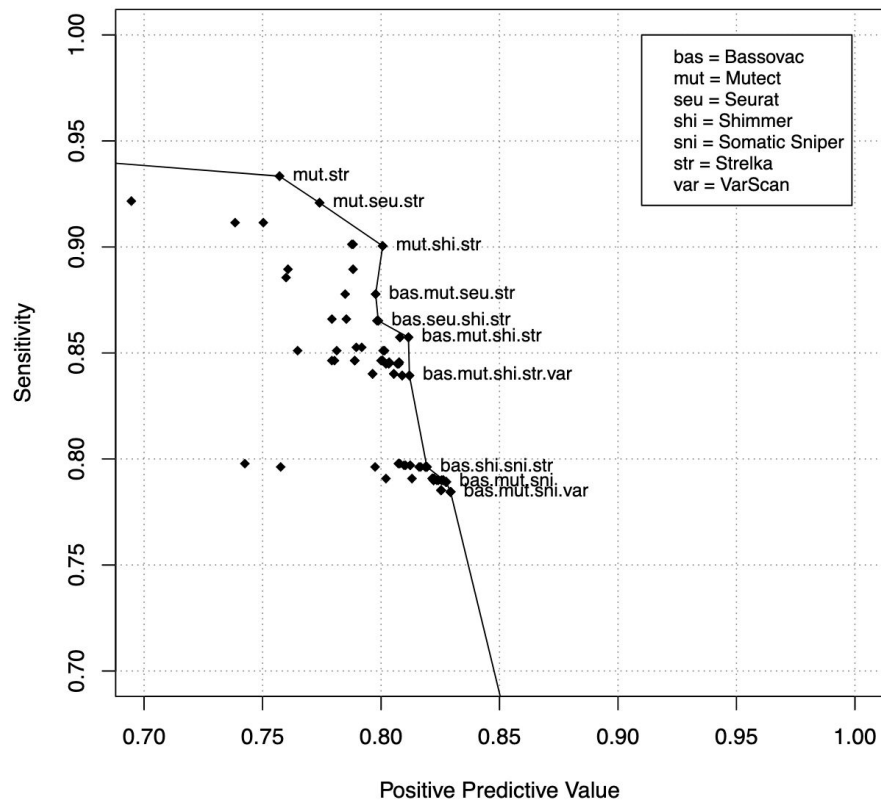


Somatic variant calling is harder

- There are more factors to consider, a wider range of possibilities, and often, more sketchy samples

Use of multiple variant callers can improve sensitivity and accuracy

Performance of caller Intersections



Somatic Variant detection workflow

- Step 1 – run the callers
 - Mutect, Strelka, VarScan, Pindel

Somatic Variant detection workflow

- Varscan

- `java -jar VarScan.v2.3.6.jar somatic <(samtools mpileup ... normal.bam) <(samtools mpileup ... tumor.bam)`

- Mutect

- `/gatk/gatk Mutect2 -O output.vcf -R ref.fa -I tumor.bam -tumor TUMORNAME -I normal.bam -normal NORMALNAME -L interval.list`

Somatic Variant detection workflow

- Step 1 – run the callers
 - Mutect, Strelka, Varscan, Pindel
- Step 2 – Merge the VCFs
 - GATK - CombineVariants

Somatic Variant detection workflow

- Combine Variants

```
/usr/bin/java -jar /opt/GenomeAnalysisTK.jar -T  
CombineVariants -genotypeMergeOptions PRIORITIZE  
--rod_priority_list mutect,varscan -o combined.vcf.gz  
--variant:mutect mutect_output.vcf --variant:varscan  
varscan_output.vcf
```

Somatic Variant detection workflow

- Step 1 – run the callers
 - Mutect, Strelka, Varscan, Pindel
- Step 2 – Merge the VCFs
 - GATK - CombineVariants
- Step 3: Annotate and filter

Assignment

- 1) Take a normal bam file and call germline variants producing a VCF file
- 2) Take a tumor and matched normal bam file and identify somatic variants using multiple callers. Produce a merged VCF with your results